



JAVA using BlueJ



An operator is a symbol that tells the compiler to perform specific mathematical or logical functions.

- Arithmetic Operators
- Relational Operators
- Logical Operators
- Assignment Operators
- Increment and Decrement Operators
- Conditional Operators
- Special Operators

Arithmetic Operators

- An arithmetic operator performs mathematical operations such as addition, subtraction and multiplication on numerical values (constants and variables).

Operator	Description	Example (int a=10,b=7;)
+	Addition	$a+b = 17$
-	Subtraction	$a-b = 3$
*	Multiplication	$a*b = 70$
/	Division	$a/b = 1$
%	Modulus (Remainder)	$a\%b = 3$

Relational Operators

- A relational operator checks the relationship between two operands.
- It returns Boolean value i.e. true or false
- Relational operators are used in decision making and loops.

Operator	Description	Example (int a=5,b=10)	Result
>	greater than	$a > b$	false
<	less than	$a < b$	true
>=	greater than or equal to	$a \geq b$	false
<=	less than or equal to	$a \leq b$	true
==	equal to	$a == b$	false
!=	not equal to	$a != b$	true

- Logical operators are used to combine two or more condition for building complex expression.
- It also return a Boolean result i.e. true or false.

Operator	Name	Description
&&	AND Operator	Checks whether the conditions preceding and succeeding it is true or not.
	OR Operator	Checks whether either of the conditions preceding and succeeding it is true or not.
!	NOT Operator	Just negates the logic of the condition succeeding it to check for its validity.

Assignment Operator

- In Java, the assignment operator (=) is used to assign a value to a variable.
 - `a=5;`
 - `x=y;`
- Chained Assignments: You can chain assignments to assign the same value to multiple variables in a single statement.
 - `b=b=c=5;`
- Compound Assignment Operators: it combine an operation and assignment in a single statement.
 - `+=`
 - `-=`
 - `*=`
 - `/=`
 - `%=`

- Write a program input 2 int and b and swap these values.
- Write a program input 2 int and b and swap these values without using third variable.

Increment/Decrement Operator

- Increment operators are used to increase the value of the variable by one and decrement operators are used to decrease the value of the variable by one.

Type of Increment Operator:

- Pre increment/decrement Operator ($++a$, $--a$)
- Post increment/decrement Operator ($a++$, $a--$)
- In pre increment / decrement first increment/decrement the value of variable and then used inside the expression (initialize into another variable)
- In post-increment / decrement first value of variable is used in the expression (initialize into another variable) and then increment / decrement the value of variable.

Conditional Operator

The conditional operator (? :) is a ternary operator. This operator needs 3 operands to perform the operation.

Syntax:

Variable = <test expression> ? <expression 1> : <expression 2>

The conditional operator works as follows:

- Test Expression is Condition
- Expression 1 is Statement Followed if Condition is True
- Expression 2 is Statement Followed if Condition is False

Precedence and Associativity of operators

Precedence:

- it determines the order in which operators are evaluated in expressions.
- Operators with higher precedence are evaluated first, followed by operators with lower precedence.

Associativity:

- If operators have the same precedence, they are evaluated from left to right (left-associativity) or from right to left (right-associativity), depending on the specific operators.

Operator	Description	Associativity
() [] . -> ++ --	Parentheses (function call) (see Note 1) Brackets (array subscript) Member selection via object name Member selection via pointer Postfix increment/decrement (see Note 2)	left-to-right
++ -- + - ! ~ (type) * & sizeof	Prefix increment/decrement Unary plus/minus Logical negation/bitwise complement Cast (convert value to temporary value of <i>type</i>) Dereference Address (of operand) Determine size in bytes on this implementation	right-to-left
* / %	Multiplication/division/modulus	left-to-right
+ -	Addition/subtraction	left-to-right
<< >>	Bitwise shift left, Bitwise shift right	left-to-right
< <= > >=	Relational less than/less than or equal to Relational greater than/greater than or equal to	left-to-right
== !=	Relational is equal to/is not equal to	left-to-right
&	Bitwise AND	left-to-right
^	Bitwise exclusive OR	left-to-right
	Bitwise inclusive OR	left-to-right
&&	Logical AND	left-to-right
	Logical OR	left-to-right
? :	Ternary conditional	right-to-left
= += -= *= /= %= &= ^= = <<= >>=	Assignment Addition/subtraction assignment Multiplication/division assignment Modulus/bitwise AND assignment Bitwise exclusive/inclusive OR assignment Bitwise shift left/right assignment	right-to-left
,	Comma (separate expressions)	left-to-right

Find the Output

- `S.O.P(4.00/(2.0+2.0))`
- `num1=2+9*((3*12)-8)/10`
`S.O.P(num1)`
- `num=24//4//2`
`S.O.P(num1)`
- `num1=20%7*2`
`S.O.P(num1)`

Type casting is process to convert a variable from one data type to another data type. For example if we want to store a integer value in a float variable then we need to typecast integer into float.

There are two type of conversion

- **Implicit:** When the type conversion is performed automatically by the compiler without programmers intervention, such type of conversion is known as **implicit type conversion**
- **Explicit:** The type conversion performed by the programmer by posing the data type of the expression of specific type is known as explicit type conversion. The explicit type conversion is also known as **type casting**.

Syntax:

(data_type) var_name;

Example:

```
int =65;      char ch=(char)a;
```

- Which among the following is a relational operator?
 - a. + b. > c. * d. /
- The && operator requires which among the following conditions to be satisfied to represent a success for two relations?
 - A. Both should be satisfied b. Any one should be satisfied.
 - c. None of them are satisfied. d. None of these
- Which among the following is not a logical operator?
 - A.&& b. || c. ! d. <=
- Which among the following operators is used to dynamically allocate memory space for an object?
 - a. . (dot) b. > c. == d. new
- Which among the following is the correct construct for creating an object named 'book' of the 'Library' class?
 - A. Library book=new Library; b. book =new Library();
 - c. Library() book=new Library(); d. Library book=new Library();

- Which operator is used to access individual members of an object?
 - a. . (dot) b. > c. == d. New
- The member 'out' object of the 'System' class belongs to which class?
 - A.System b. Print c. PrintStream d. Println
- The statement `a%2` will result to what if the value of `a` is an even number?
 - a. 0 b. 1 c. Both a and b d. None of these
- Which among the following operators is used to check the two operands are equal or not?
 - a. Equals b. = c. == d. None of these
- Which among the following forms of operators works with three operands?
 - a. Unary b. Binary c. Ternary d. None of these